

Probabilistic Machine Learning

Lecture 05: Bayesian Networks

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Belief networks in depth, plus the two topics requested for this lecture: fitting (parameters and structure) and graphical notation. The plate-notation unit is new everything else is reused. Markov networks are included as a contrast — drop that module for a shorter course.

Module 1

Belief Networks

Guiding Question: *What exactly does an arrow in a probabilistic model claim — and what does a missing arrow claim?*

Definition

Bayesian belief network (BBN). A **directed acyclic graph** whose nodes are random variables, in which the joint distribution factorises as

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i \mid \text{pa}(x_i))$$

where $\text{pa}(x_i)$ are the parents of x_i in the graph.

Each node carries one conditional probability table given its parents. The wet-grass model was exactly this.

Math first (P5) — the students have already built one of these by hand, so the general form should read as a summary rather than a novelty. Stress "acyclic": cycles would make the factorisation ill-defined.

Three Tasks

Modelling

Choose variables, edges and tables that encode the domain.

Inference

Compute posteriors given evidence.
Exact **and** approximate inference are **NP-hard** in general.

Learning

Estimate the tables — and possibly the structure — from data.

The NP-hardness is worth dwelling on: tractability depends on the *shape* of the network, not on its size alone. Trees are easy, dense graphs are not.

Modelling Technique: Divorcing Parents

A node with k binary parents needs 2^k table entries. **Divorcing** inserts an intermediate variable that summarises a group of parents, so the table becomes two small ones instead of one large one.

The intermediate node usually has a meaning of its own — "total load", "combined risk" — which is why the technique improves interpretability as well as size.

Fewer parameters means fewer numbers a domain expert has to invent.

Frame parameter reduction as an elicitation problem, not just a storage one: an expert can estimate 4 numbers reliably and 64 numbers not at all.

Modelling Technique: Noisy OR

When many causes $x_1, \dots, x_k$ can each independently trigger an effect y , a **noisy OR** models the effect as "each cause fires with its own probability, and the effect occurs if any of them fires":

$$p(y = 0 \mid x_1, \dots, x_k) = \prod_{i: x_i=1} (1 - \lambda_i)$$

k parameters instead of 2^k .

Typical use

Disease–symptom networks, where many diseases give rise to the same symptom.

Ask what assumption buys the reduction: causes act independently and there is no interaction term. Then ask when that fails — synergistic causes — so the model choice stays a modelling decision.

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

The state of x is itself uncertain, expressed as a distribution $\tilde{x}$. Infer by summing over the states:

$$p(y \mid \tilde{x}) = \sum_x p(y \mid x)p(x \mid \tilde{x})$$

We look at the grass without touching it:

$$p(T=1 \mid \tilde{T}) = 0.8.$$

Unreliable evidence

The evidence is not trusted and is **replaced** by our own estimate:

$$p(y \mid x) \rightarrow p(h \mid x)$$

Jack says his grass is wet, but Jack is in a bad mood and did not check.

Both are drawn with a **dashed** edge.

The distinction is subtle and worth a moment: uncertain evidence keeps the observation and softens it unreliable evidence discards it and substitutes a judgement. Students who conflate them will model rumours as noise.

Takeaway

A belief network is a **DAG plus one conditional table per node**. Modelling techniques — divorcing, noisy OR, soft evidence — all trade assumptions for parameters, and every trade should be stated out loud.

Module 2

Reading Independence off the Graph

*Guiding Question: Two causes are independent until you observe their shared effect.
Which pictures make that visible?*

Three Ways Three Nodes Can Connect

Every conditional-independence question in a DAG reduces to these three patterns.

Chain

$$a \rightarrow c \rightarrow b$$

a and b are dependent — but **independent given c** .

Information flows along the chain until the middle is observed.

Common cause (fork)

$$a \leftarrow c \rightarrow b$$

a and b are dependent — but **independent given c** .

The shared cause explains the correlation.

Common effect (collider)

$$a \rightarrow c \leftarrow b$$

a and b are **independent** — but **dependent given c** .

Explaining away — the reverse of the other two.

The whole slide is one table students should be able to reproduce. Emphasise the collider's inversion: conditioning normally destroys dependence, here it creates it. The wet-grass example is exactly a collider at T.

Mini-Exercise

For each pair, decide whether observing the middle variable makes the outer two *more* or *less* related:

1. rain $\rightarrow$ wet grass $\rightarrow$ slippery pavement, observing wet grass
2. temperature $\rightarrow$ ice-cream sales, temperature $\rightarrow$ drownings, observing temperature
3. rain $\rightarrow$ wet grass $\leftarrow$ sprinkler, observing wet grass

Mini-Exercise — Solution

Solution fragmented per the animation policy. Case 3 is the one that surprises; if the group is quick, ask what happens when you condition on a *descendant* of the collider (the same dependence appears — which is why d-separation mentions descendants).

Mini-Exercise — Solution

1. Chain

Independent given wet grass
— the pavement only cares
that the ground is wet.

2. Fork

Independent given
temperature — the classic
spurious correlation dissolves.

3. Collider

Dependent given wet grass —
learning it rained lowers belief
in the sprinkler.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains either

- a chain or fork whose middle node **is in** Z , or
- a collider whose middle node — and all of its descendants — are **not in** Z .

If every path from X to Y is blocked by Z , then X and Y are **d-separated** given Z , and $X \perp \perp Y \mid Z$.

d-separation is sound: what it declares independent really is independent for every distribution factorising over the graph. It is not complete for a *particular* distribution — extra independences may hold numerically without being visible in the graph.

Definition after the three cases, so it reads as a summary rather than an axiom. The soundness/completeness caveat is the honest version of "graphs have limited expressive power" from the intro unit.

Takeaway

Independence in a belief network is a **graph property**: chains and forks are blocked *by* observing the middle node, colliders are opened by it. d-separation is that rule applied to every path.

Module 3

Graphical Notation

Guiding Question: *A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn?*

Drawing a Model

A probabilistic model is a factorisation, and the graph is its picture. Four conventions carry almost all the information:

Shaded node

An **observed** variable — its value is in the data.

Unshaded node

A **latent** variable or a parameter — inferred, never seen.

Small solid dot

A fixed **hyperparameter**, not inferred.

Arrow

A conditional dependence: the child's distribution is conditioned on the parent.

Insist that this is notation, not new theory: the diagram is a compact way to write a factorisation students can already read. The shading convention is what they will most often need when reading papers.

Plates

A **plate** is a rectangle drawn around part of the graph with a count in its corner: *repeat this structure N times, with independent copies of the variables inside.*

It is the i.i.d. assumption, drawn. Without it, a model of N observations would need N copies of the same subgraph.

Nested plates express grouped data — N documents each of M_n words, or K hospitals each with N_k patients.

What sits outside the plate

Anything **shared** across observations — the parameters. That is precisely what learning estimates.

Reading a Diagram

For a Bayesian linear regression: a hyperparameter dot feeds $\mathbf{w}$, which sits **outside** the plate inside a plate over N observations sit $\mathbf{x}_i$ (shaded) and y_i (shaded), with $\mathbf{w}$ pointing into each y_i .

Read off the factorisation directly: $p(\mathbf{w} \mid \tau) \prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w})$. The plate became the product the arrows became the conditioning bars.

Mini-Exercise

A diagram shows a dot $\rightarrow \alpha \rightarrow \theta$ inside a plate of K , and $\theta \rightarrow z_i \rightarrow x_i$ inside a plate of N , with x_i shaded.

1. Which variables are inferred?
2. What does the N plate assert?
3. Write the factorisation.

Mini-Exercise — Solution

Plate notation compresses a factorisation into a diagram: **shading** marks what was observed, **plates** mark what repeats independently, and **what sits outside the plates** is what learning has to estimate.

Part 3 is the skill worth having: converting a diagram into a formula and back. Every probabilistic-modelling paper the students read assumes it.



Mini-Exercise — Solution

1. θ and every z_i — unshaded. Only x_i is observed α is fixed.

2. The N observations are **conditionally independent given θ** — each with its own latent z_i .

3. $p(\theta \mid \alpha) \prod_{i=1}^N p(z_i \mid \theta) p(x_i \mid z_i)$ — a mixture model.

Plate notation compresses a factorisation into a diagram: **shading** marks what was observed, **plates** mark what repeats independently, and **what sits outside the plates** is what learning has to estimate.

Module 4

Undirected Models

Guiding Question: *Pixels influence each other symmetrically — no pixel causes its neighbour. How do we model that?*

Undirected Models

Markov network (Markov random field). An **undirected** graph in which the joint distribution factorises over the **cliques** $c \in \mathcal{C}$ of the graph:

$$p(x_1, \dots, x_n) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \phi_c(x_c), \quad Z = \sum_x \prod_c \phi_c(x_c)$$

The $\phi_c \geq 0$ are **potentials** — arbitrary non-negative functions, not probabilities. Z is the normalising **partition function**.

When all cliques have size 2 the model is a **pairwise Markov network**.

The key difference from a BBN: potentials are not conditional probabilities and do not normalise locally, so a global Z is required. Computing Z is where the cost lives.

Directed vs. Undirected

Belief network

Directed, acyclic. Factors are conditional probabilities. Natural for **causal or generative** stories conditioning on a collider creates dependence.

Markov network

Undirected. Factors are potentials with a global Z . Natural for **symmetric interactions** — pixels, atoms, neighbours in a lattice.

Neither subsumes the other: some independence patterns are expressible only in one. Give the standard examples if time permits (a collider is not expressible undirected a 4-cycle is not expressible directed).

Marginalising and Conditioning Behave Oppositely

Marginalising over C

makes A and B **dependent** — the sum over all states of c couples the potentials.

Conditioning on C

In the chain $A - C - B$, a fixed c is a constant in the potentials, so the factor splits and $A \perp\!\!\!\perp B \mid C$.

This holds because C separates them — not for every C .

This pair is the undirected counterpart to chains, forks and colliders. In an undirected graph the rule is simple: separation *is* conditional independence, no collider exception.

Markov Properties and the Separation Test

For undirected models, independence is plain graph separation: if every path from X to Y passes through Z , then $X \perp\!\!\!\perp Y \mid Z$.

For a **directed** model the same question is answered by converting the graph:

1. Ancestral graph

Keep $X \cup Y \cup Z$ and their ancestors delete all other nodes and their edges.

2. Moralisation

Connect any two nodes that share a child, then drop all arrow heads.

3. Separation

Remove the links neighbouring Z . If no path joins X to Y , then $X \perp\!\!\!\perp Y \mid Z$.

"Marrying the parents" is where moralisation gets its name, and it is exactly the collider rule made structural: parents of a common child become connected, so they are no longer separable. This procedure is equivalent to d-separation.

Two Examples

Boltzmann machine

A pairwise Markov network over binary units with weights on the edges the restricted variant (RBM) allows edges only between a visible and a hidden layer.

Image cleanup

Pixels form a lattice: each potential prefers a pixel to agree with its neighbours and with the noisy observation. Inference denoises the image.

The image example is the most intuitive Markov network there is — smoothness as a potential. It also previews the energy-based view used for neural networks later.

Other Graphical Models

Chain graphs

Combine directed and undirected edges in one model.

Factor graphs

Make each factor an explicit node — mostly used **for inference**, since they state precisely which variables each factor touches.

Undirected models trade causal direction for symmetric interaction, and local normalisation for a global partition function. Separation in the (moralised) graph is the single test for conditional independence.

Module 5

Fitting and Inference

Guiding Question: *We can now write the model down. Can we actually compute anything with it?*

Three Questions You Can Ask a Model

Infer unobserved variables

Compute the posterior given evidence — probabilistic inference.

Learn the parameters

Estimate the factors' distributions from data.

Learn the structure

Discover which dependencies exist at all.

The three get progressively harder — and progressively more interesting.

Structure learning is where causal discovery lives flag it as an open research area rather than a solved technique, since students will meet confident claims about it.

Inference Is Hard in General

Both exact and approximate inference in belief networks are **NP-hard**. Whether a particular model is tractable depends on its structure.

Exact inference

Variable elimination, junction trees, message passing.
Efficient when the graph is a tree or has low treewidth.

Approximate inference

Sampling (MCMC, importance sampling) and variational methods.
Trade a guarantee for tractability.

The practical rule: structure decides. A chain or tree is linear a dense graph is hopeless. This is why modelling and inference are not separable concerns — a model you cannot query is not a model.

Learning the Parameters

With **fully observed** data, parameter learning is counting: each conditional table is estimated from the frequencies of its variable and parents.

With **hidden variables**, the counts are unavailable and one alternates between inferring the hidden values and re-estimating the parameters (expectation–maximisation).

The estimates are only as good as the sample supporting each table row. A parent combination that never occurs in the data leaves its row undetermined — which is another argument for keeping tables small.

Connect back to divorcing and noisy OR: parameter reduction is also a statistical necessity, not only a storage convenience. This also foreshadows smoothing in naive Bayes.



Takeaway

Graphical models make the joint distribution **writable** whether they make it **queryable** depends on the graph's structure. Parameters come from counting where data is complete, and from iterating where it is not.